Bayesian Model Selection

Pick two alternative descriptions of the data,

M_0

M_1

$\Rightarrow$ Calculate Bayesian evidence for each,

$p(d|M_0)$ → ratio gives us the

$p(d|M_1)$ Bayes factor

$\frac{p(d|M_1)}{p(d|M_0)}$ Bayes factor	interpretation
$<10^0$	prefer M_0
$10^0 \sim 10^{1/2}$	"barely worth mentioning"
$10^{1/2} \sim 10^1$	"substantial"
$10^1 \sim 10^{3/2}$	strong
$10^{3/2} \sim 10^2$	very strong
$>10^2$	descisive

Jeffrey's scale

Theoretically, to calculate evidence;

$$\int \underbrace{p(d|\theta, M)}_{\text{likelihood}} \cdot \underbrace{p(\theta|M)}_{\text{prior}} \, d\theta = \underbrace{p(d|M)}_{\text{evidence}}$$

Laplace Approximation

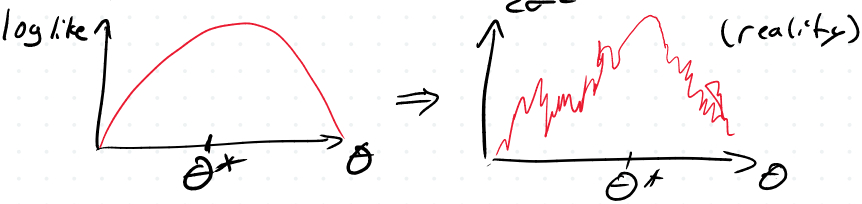
Approximation of likelihood around the best-fit value. Maximum likelihood estimate (MLE)

$\theta^* \Leftarrow$ parameters that maximize $p(d|\theta)$

Assume that likelihood is Gaussian around the peak.

$p(d|\theta) \propto e^{-(\theta-\theta^*)^2/2\sigma^2}$ (something like this)

$\log p(d|\theta)$ is quadratic $\sim \frac{1}{2\sigma^2}(\theta-\theta^*)^2$



$$\log p(d|\theta) = \log p(d|\theta^*) + (\theta^i - \theta^{*i}) \underbrace{\partial_i \log p(d|\theta)}_{\to 0 \text{ b/c max}}$$

(1)

$$+ \frac{1}{2}(\theta^i - \theta^{*i})(\theta^j - \theta^{*j}) \underbrace{\partial_i \partial_j \log p(d|\theta)}_{\text{Hessian matrix } H_{ij}}$$

$$+ \mathcal{O}(\theta^3)$$

so:

$$\int d\theta \, p(d|\theta, M) \simeq \int \underbrace{p(d|\theta^*, M)}_{\text{constant}} \exp\left(-\frac{1}{2}(\theta - \theta^*)^2_{ij} H^{ij}\right) d\theta$$

$$= p(d|\theta^*, M) \int d\theta \, e^{-\frac{1}{2}(\theta - \theta^*)_i H_{ij} (\theta - \theta^*)_j}$$

$$= p(d|\theta^*, M) \, (2\pi)^{D/2} \, |H|^{-1/2}$$

$$\boxed{p(d|M)_L = \underbrace{p(d|\theta^*, M)}_{\text{max. like}} \, \underbrace{p(\theta^*|M)}_{\text{prior @ max. like}} \cdot \frac{(2\pi)^{D/2}}{|H|^{1/2}}}$$

Sometimes called Laplace-Fisher approximation.

Fisher approximation for Likelihood:

This is basically just (1).

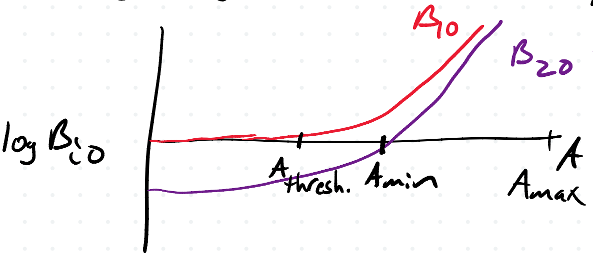
How to think about it:

- Max. likelihood — bonus to evidence for a good fit
- prior — 1/V cost for extra degrees of freedom
- gauss. integral — discounts unconstrained parameters

"Occam factor"

evidence ~ goodness of fit × occam factor

Ex/ M_0: noise
M_1: signal w/ amplitude A, $p(A) \sim U[0, A_{max}]$
M_2: signal w/ amplitude A, $p(A) \sim U[A_{min}, A_{max}]$



$$B_{20} = \frac{\text{evidence } M_2}{\text{evidence } M_0} = \frac{p(d|M_2)}{p(d|M_0)}$$

As $A \longrightarrow 0$, we can rule out M_2, but never M_1.

Bayesian Information Criterion (BIC)

- Same as Laplace-Fisher approx., but we assume we have infite sample size.

$$\underbrace{\log p(d|M)_{BIC}}_{} = \underbrace{\log l(d|\theta^*, M)}_{\text{goodness of fit}} - \underbrace{\frac{D}{2}\log N}_{\text{penalty for \# params (D) and for lots of data.}}$$

$$AIC \equiv -2\log(p(d|\theta^*)) + 2D$$

$$DIC \equiv -2D_{KL} + 2C_D \rightarrow \text{effective \# params}$$

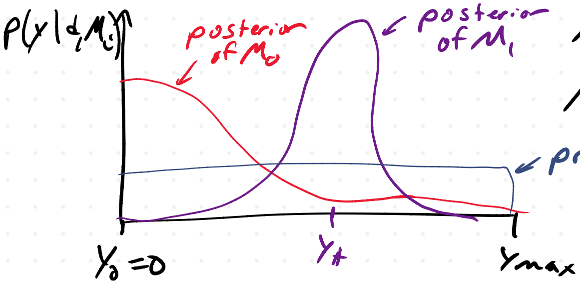
Akaike information criteria

Savage-Dickey Density

Works for nested models $\hookrightarrow$ M_0 nests inside M_1 if $M_1|\theta_i = \bar{\theta}_i = M_0$.

fix this param., then $M_1 = M_0$

$$B_{10} = \frac{p(\theta|M_1, \theta_i = \bar{\theta}_i)}{p(\theta|M_0)}$$



$M_0 : \theta \rightarrow \{x, y = y_0\}$

$M_1 : \theta \rightarrow \{x, y\}$

$$B_{10} = \frac{p(y=y_0 \mid d, M_1)}{p(y=y_0 \mid d, M_0)}$$

Hierarchical Analysis

Maybe my sources all come from a population, what describes that population?

Population is described by <u>hyperparameters</u>.

For one source, we have:

$$p(\theta \mid d) = \frac{p(d \mid \theta)\, p(\theta)}{p(d)}$$

But we have hyperparameters ϕ.

$$p(\theta, \phi \mid d) = \frac{p(d \mid \theta)\, p(\theta, \phi)}{p(d)}$$

$$p(d) = \int p(d \mid \theta)\, p(\theta, \phi)\, d\theta\, d\phi$$

e.g.

1) Lots of BBHs, we want to know the mass spectrum and merger rate.
2) spatial distribution of galactic sources
3) Alternative theory of gravity
   Brans–Dickie theory
   Gravity's rainbow

Expand the prior $p(\theta|\varphi)\,p(\varphi)$

hyperprior

$$p(\theta,\varphi|d) = \frac{p(d|\theta)\,p(\theta|\varphi)\,p(\varphi)}{p(d)}$$

$$p(\varphi|d) = \int p(\theta,\varphi|d)\,d\theta$$ ← posterior on hyperparameters only!

$$p(\theta|d) - \int p(\theta,\varphi|d)\,d\varphi$$ ← posterior of one source.

You'd have to know the population to get a good estimate of one source.

⇒ Next time, finish hierarchical models, stochastic sampling algorithms.